

Economic Prosperity and Football Success: A Comparative Study of Turkey & England (1993–2024)

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Motivation:

Turkish football history features iconic peaks — Galatasaray's 2000 UEFA Cup and Super Cup, Beşiktaş becoming the first and only Turkish club to top their Champions League group stage unbeaten (2017–18), and the national team's historic 3rd-place finish at the 2002 World Cup. Each of these achievements coincided with periods of relative macroeconomic stability in Turkey, raising a core question: does economic health act as a catalyst for international football success? This project investigates whether macroeconomic indicators — exchange rates, GDP growth, and CPI inflation — are statistically associated with Turkish clubs' UEFA coefficient performance and the national team's FIFA ranking. England's Premier League Big 6 and national team are incorporated as a structural counterpoint, enabling a cross-country comparison and increasing statistical power for machine learning.

Data Sources & Collection:

Two categories of data are used:

- **Economic Indicators:** USD/TRY half-year average exchange rates from the Central Bank of Turkey (TCMB EVDS); GBP/USD from the Bank of England; annual GDP growth rates and CPI inflation for both Turkey and England from World Bank Open Data (indicators NY.GDP.MKTP.KD.ZG and FP.CPI.TOTL.ZG).
- **Football Performance:** Per-club half-year UEFA coefficient points for Galatasaray, Fenerbahçe, Beşiktaş, and Trabzonspor (Turkey) and the Premier League Big 6 (England), reconstructed from UEFA.com and kassiesa.net/uefa/coeff; bi-annual FIFA World Ranking snapshots for both national teams from FIFA's historical archive.

Data Characteristics:

The dataset covers 31 years (1993–2024) with half-year granularity (H1: January–June; H2: July–December), yielding 64 observations per country and a combined dataset of 128 rows for machine learning. A composite Economic Health Index (0–1 scale) is constructed by min-max normalising and averaging the three economic indicators, with exchange rate and inflation inverted so that higher index = better economic conditions.

Methodology:

The analysis follows four phases. First, exploratory data analysis with time-series visualisations and dual-axis overlays of economic and football metrics. Second, formal hypothesis testing using one-tailed Pearson correlation ($H_1: \rho > 0, \alpha = 0.05$) and Spearman ρ as a non-parametric backup, with D'Agostino K^2 normality checks. Third, OLS regression of total club UEFA points on the three economic predictors jointly, to quantify combined explanatory power (R^2). Fourth, machine learning: K-Means clustering (elbow method) and PCA for unsupervised era identification, and three supervised classifiers (Logistic Regression, Random Forest, Gradient Boosting) with 5-fold stratified cross-validation to classify high- versus low-performance periods from economic features alone. All code is written in Python using pandas, scipy, statsmodels, and scikit-learn.